MakeNoise<DI, MI, MO> for DiscreteGaussian

Michael Shoemate

Proves soundness of the implementation of `MakeNoise` for `DiscreteGaussian` in `mod.rs` at commit f5bb719 (outdated[1]).

This is an intermediary compile-time layer whose purpose is to dispatch to either the integer or floating-point variations of the mechanism, depending on the type of data in the input domain.

It does this through the use of the `Nature` trait, which has concrete implementations for each possible input type. This layer makes interior layers simpler to work with, and does not have privacy implications. It also makes `make_gaussian` easier to call, by simplifying the type signature.

1 Hoare Triple

Precondition

Compiler-Verified

`MakeNoise` is parameterized as follows:

- `DI` implements trait `Domain`
- `MI` implements trait `Metric`
- `MO` implements trait `Measure`

The following trait bounds are also required:

- `(DI, MI)` implements trait `MetricSpace`
- `DI_Atom` implements trait `Nature`. This trait encodes the relationship between the atomic data type and the type of the noise distribution that is compatible with it: `DI_Atom_RV2`. In Rust, this corresponds to the (ugly) `<DI::Atom as Nature>::RV<2>` type.
- `DI_Atom_RV2` implements trait `MakeNoise`. That is, it must be possible to build the mechanism from this new equivalent distribution.

User-Verified

None

Pseudocode

```python
# analogous to impl MakeNoise<DI, MI, MO> for DiscreteGaussian in Rust
class DiscreteGaussian:
    def make_noise(self, input_space: tuple[DI, MI]) -> Measurement[DI, DI_Carrier, MI, MO]:
        # an equivalent random variable specific to the atom dtype
        rv_nature = DI_Atom.new_distribution(self.scale, self.k) #
        # build a measurement sampling from this equivalent distribution
        return rv_nature.make_noise(input_space) #
```

[1] See new changes with `git diff f5bb719..31c8a35f rust/src/measurements/noise/distribution/gaussian/mod.rs`

Postcondition

Theorem 1.1. For every setting of the input parameters (`self, input_space, DI, MI, MO`) to `make_noise` such that the given preconditions hold, `make_noise` raises an error (at compile time or run time) or returns a valid measurement. A valid measurement has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Privacy guarantee). For every pair of members x and x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
`output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map`(`d_in`) does not raise an error, and `privacy_map`(`d_in`) = `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_measure`.

Proof. On line 7, `make_noise` has no preconditions, so irregardless of any prior logic, the postcondition of `make_noise` follows that the output is a valid measurement. □

The complexity in the type system here is designed to be free of privacy implications, to help simplify the core, privacy-sensitive implementation.